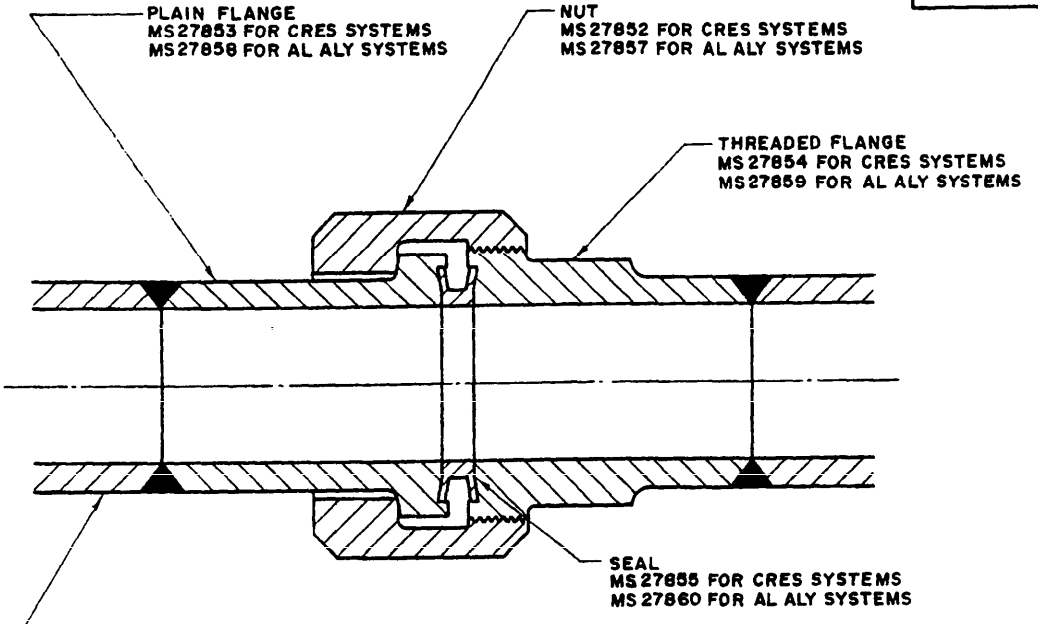
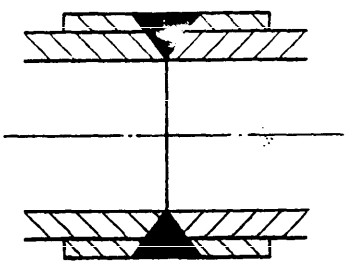
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OPTIONAL JOINT DESIGN

TUBING TO TUBING FITTING INSTALLATION

DASH NO. REF.		WRENCH TORQUE FOR TIGHTENING CRES FITTINGS (LB/IN) MS27851		WRENCH TORQUE FOR TIGHTENING AL ALY FITTINGS (LB/IN) MS27856	
		MINIMUM	MAXIMUM	MINIMUM	MAXIMUM
-02	1/8	75	90		
-04	1/4	200	225	100	125
-06	3/8	380	435	175	200
-08	1/2	490	555	280	320
-12	3/4	1240	1400	530	600
-16	1	2300	2580	1070	1200

Review activities: 11, 14, 19, 69, 85 & NAS

APPROVED 14 JULY 1967 REVISED

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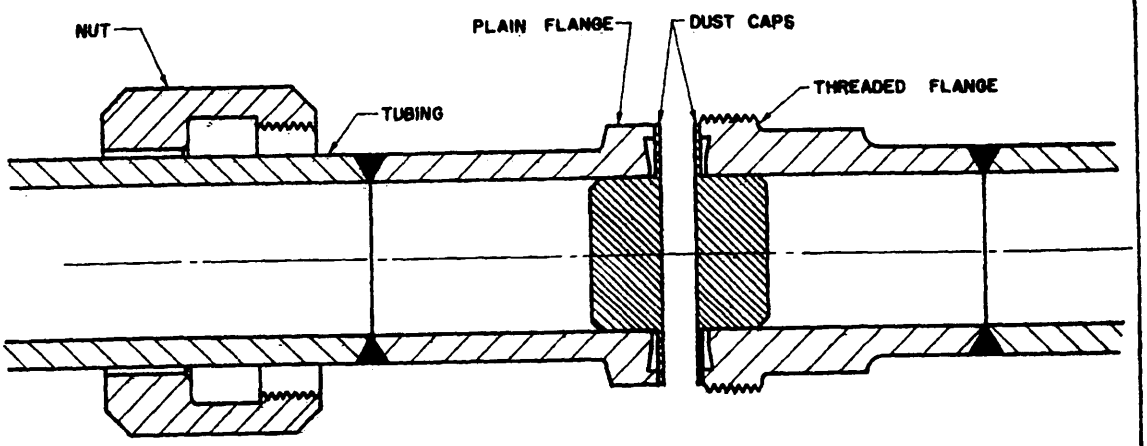
P.A. USAF 12 Other Cust	TITLE FITTINGS, INSTALLATION OF STRAIGHT THREADED FLUID CONNECTION	MILITARY STANDARD MS 27850 (USAF)
PROCUREMENT SPECIFICATION MIL-F-27417	SUPERSEDES:	SHEET 1 OF 5

DD FORM 1 SEP 67 672-1 (Limited coordination)

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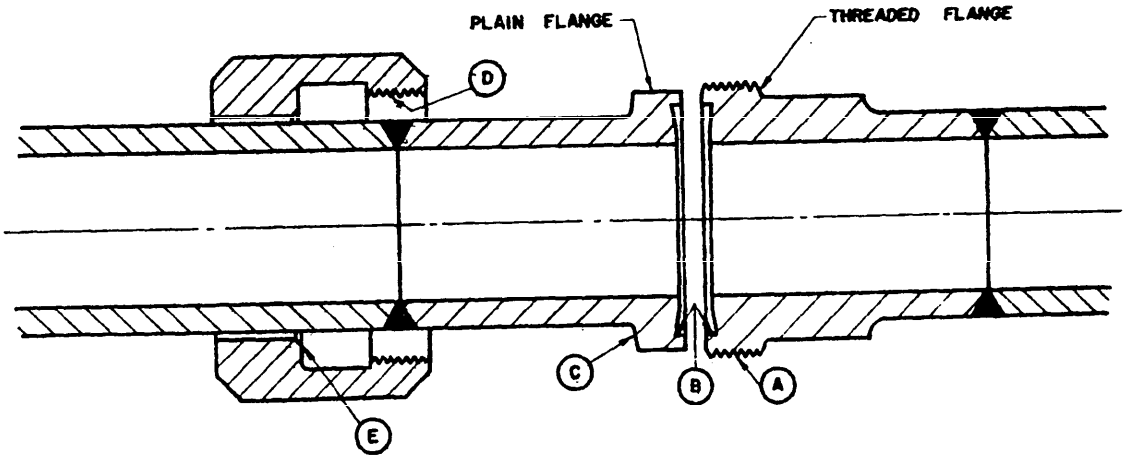
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STEP 1

PLACE THE NUT ON THE TUBING TO BE ATTACHED TO PLAIN FLANGE. WELD PLAIN AND THREADED FLANGES TO TUBING.



STEP 2

REMOVE DUST CAPS. INSPECT SURFACES A, B, C, D, AND E FOR DIRT, CONTAMINATION, OR SURFACE IRREGULARITIES. IF NECESSARY RINSE SURFACES WITH METHYLETHYLKETONE AND DRY PARTS WITH FILTERED AIR STREAM OR WIPE PARTS WITH CLEAN PAPER TISSUE MOISTENED WITH METHYLETHYLKETONE. REPLACE DUST CAPS.

Review activities: 11, 14, 19, 69, 85, 8 NAS

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APPROVED 14 JULY 1967 REVISED

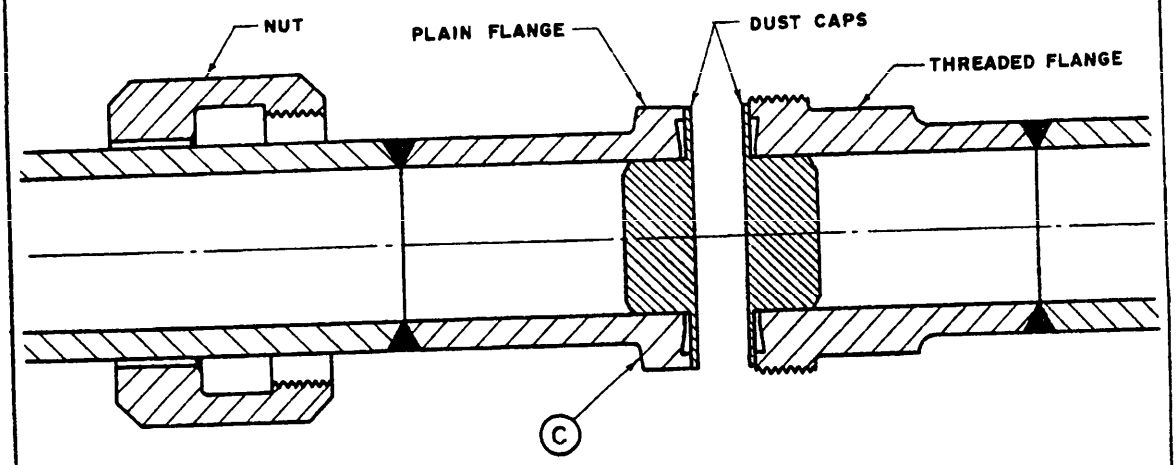
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PROCUREMENT SPECIFICATION MIL-F-27417	SUPERSEDES:	SHEET 2 OF 6

DD FORM 672-1 (Limited coordination)

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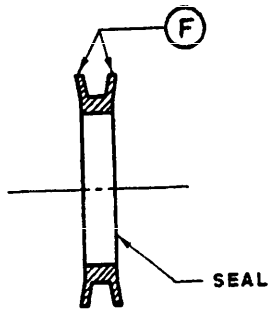
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STEP 3

COAT MALE THREADS AND SURFACE C WITH LUBRICANT. APPLY LUBRICANT SPARINGLY. WORK THE NUT ONTO THE MALE THREADS AND AGAINST SURFACE C BY HAND. RETRACT NUT AND INSPECT SURFACES FOR EVEN COATING OF LUBRICANT. REMOVE EXCESS LUBRICANT FROM NONLUBRICATED SURFACES.



STEP 4

REMOVE SEAL FROM PROTECTIVE WRAPPING. INSPECT SURFACES F WITH 10 POWER LENS. DISCARD IF SURFACES ARE SCRATCHED OR IRREGULAR. CLEAN WITH METHYLETHYLKETONE IF DIRTY.

APPROVED 14 JULY 1967 REVISED

Review activities: 11,14,19,69,85 ■ NAS

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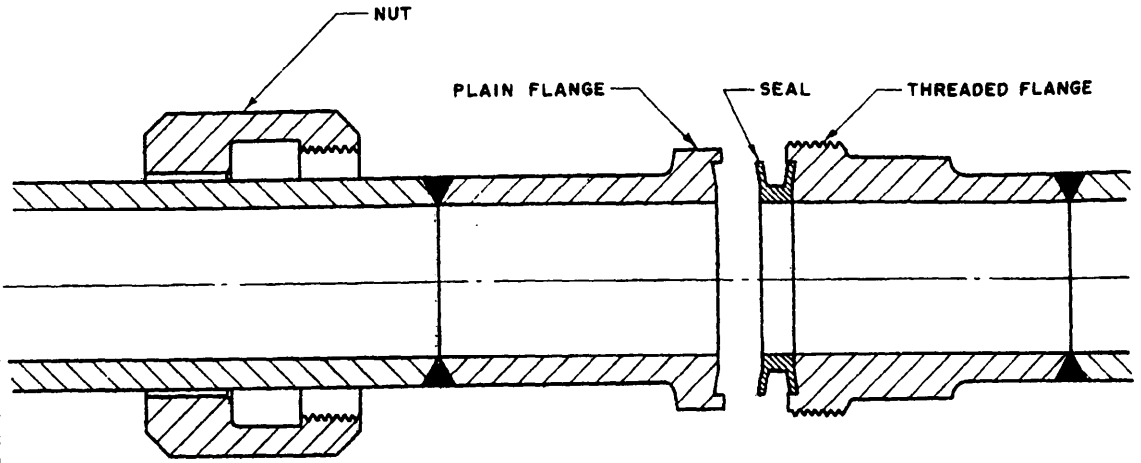
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PROCUREMENT SPECIFICATION MIL-F-27417	SUPERSEDES:	SHEET 3 OF 5

DD FORM 672-1 (Limited coordination)

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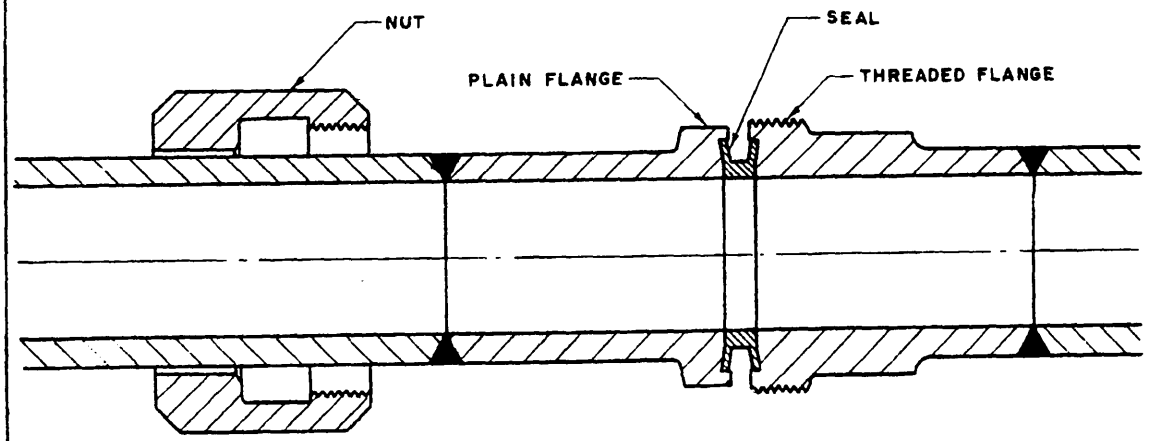
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STEP 5

REMOVE DUST CAPS FROM PLAIN AND THREADED FLANGE AND INSERT SEAL. HANDLING WITH CLEAN HANDS IS PERMISSIBLE BUT CARE MUST BE EXERCISED TO AVOID GETTING LUBRICANT OR DIRT ON THE SEALING SURFACES.



STEP 6

LOCATE PLAIN FLANGE SO SEAL IS RETAINED IN THE SEAL CAVITY OF BOTH FLANGES. MAKE CERTAIN SEAL IS COMPLETELY INSERTED INTO BOTH PLAIN AND THREADED FLANGE SEAL CAVITIES.

APPROVED 14 JULY 1967 REVISED

Review activities: 11, 14, 19, 69, 85 & NAS

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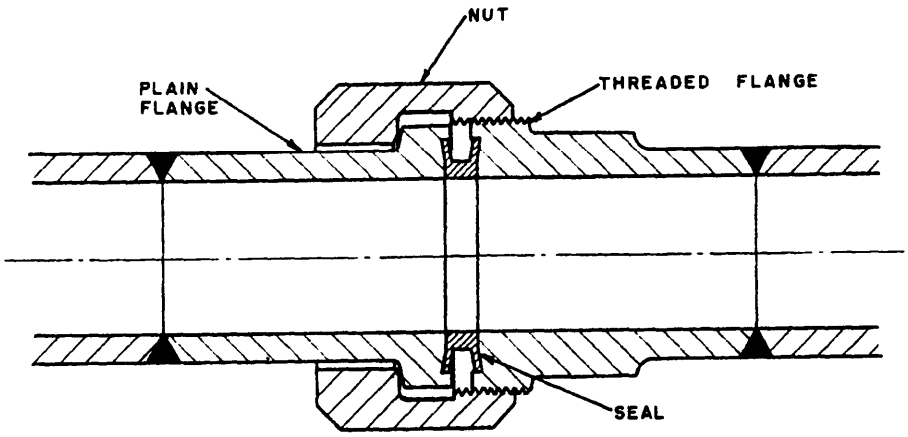
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DD FORM 1 SEP 67 672-1 (Limited coordination)

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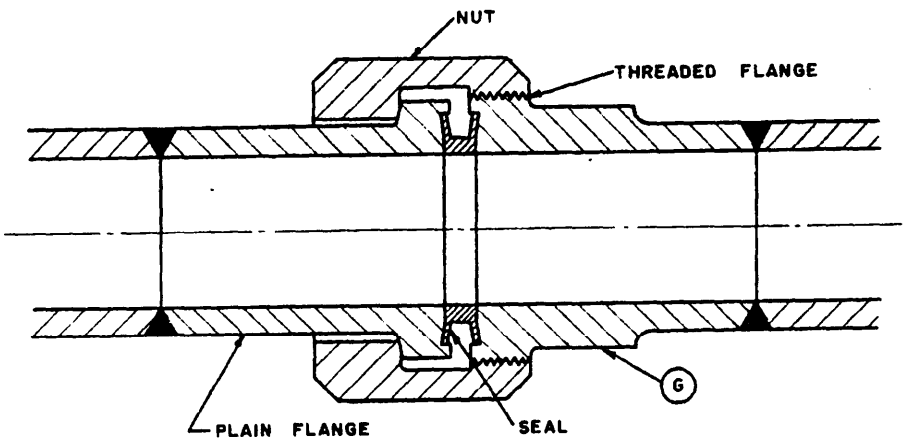
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STEP 7

TIGHTEN NUT HAND-TIGHT WHILE MAINTAINING LIGHT, FIRM PRESSURE ALONG AXIS OF FITTING TO PREVENT SEAL FROM SLIPPING OUT OF FLANGE CAVITIES.



STEP 8

PLACE OPEN-END WRENCH ON THREADED FLANGE, SURFACES G AND HOLD OR ALLOW TO BEAR AGAINST RIGID STRUCTURE TO PREVENT ROTATION. TIGHTEN NUT WITH CALIBRATED TORQUE WRENCH TO RECOMMENDED TORQUE. IF EITHER FLANGE HAS RELATIVE ROTATION MORE THAN 1/16 TURN, DISASSEMBLE AND REPLACE SEAL. PROPER ASSEMBLY IS INDICATED BY THREAD ENGAGEMENT. NO MORE THAN ONE THREAD OF EITHER THE THREADED FLANGE OR NUT SHOULD BE EXPOSED.

APPROVED 14 JULY 1967 REVISED

Review activities: 11, 14, 19, 69, 85, & NAS

THIS STANDARD HAS BEEN APPROVED BY THE DEPARTMENT OF THE AIR FORCE AND IS MANDATORY FOR USE BY THAT ACTIVITY. ALL OTHER MILITARY ACTIVITIES ARE REQUIRED TO EMPLOY THIS STANDARD WHERE SUITABLE

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PROCUREMENT SPECIFICATION MIL-F-27417	SUPERSEDES:	SHEET 5 OF 5

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